

## Lab Practical #10:

### 1. Implementation of AES Algorithm.

#### Program:

```
// 1. Implementation of AES Algorithm.

#include <stdio.h>

#define Nr 10

char state[4][4];

// Simple S-box (short version for lab)
char sbox[16] = {0x6,0x4,0xC,0x5,0x0,0x7,0x2,0xE,0x1,0xF,0x3,0xD,0x8,0xA,0x9,0xB};

void AddRoundKey(char key[4][4])
{
    for(int i=0;i<4;i++)
        for(int j=0;j<4;j++)
            state[i][j] ^= key[i][j];
}

void SubBytes()
{
    for(int i=0;i<4;i++)
        for(int j=0;j<4;j++)
            state[i][j] = sbox[state[i][j] % 16];
}

void ShiftRows()
{
    char temp;

    temp = state[1][0];
    state[1][0]=state[1][1];
    state[1][1]=state[1][2];
    state[1][2]=state[1][3];
    state[1][3]=temp;

    char t1 = state[2][0];
    char t2 = state[2][1];
    state[2][0]=state[2][2];
    state[2][1]=state[2][3];
    state[2][2]=t1;
    state[2][3]=t2;
```

```
temp = state[3][3];
state[3][3]=state[3][2];
state[3][2]=state[3][1];
state[3][1]=state[3][0];
state[3][0]=temp;
}

void MixColumns()
{
    for(int i=0;i<4;i++)
    {
        char t = state[0][i]^state[1][i]^state[2][i]^state[3][i];
        state[0][i]^=t;
        state[1][i]^=t;
        state[2][i]^=t;
        state[3][i]^=t;
    }
}

void printState()
{
    for(int i=0;i<4;i++)
    {
        for(int j=0;j<4;j++)
            printf("%d ",state[i][j]);
        printf("\n");
    }
}

int main()
{
    char plaintext[4][4];
    char key[4][4];

    printf("Enter 16 plaintext values:\n");
    for(int i=0;i<4;i++)
        for(int j=0;j<4;j++)
            scanf("%d",&plaintext[i][j]);

    printf("Enter 16 key values:\n");
    for(int i=0;i<4;i++)
        for(int j=0;j<4;j++)
            scanf("%d",&key[i][j]);

    for(int i=0;i<4;i++)
```

Date: 13/03/2026

```
        for(int j=0;j<4;j++)
            state[i][j]=plaintext[i][j];

        AddRoundKey(key);

        for(int round=1; round<Nr; round++)
        {
            SubBytes();
            ShiftRows();
            MixColumns();
            AddRoundKey(key);
        }

        SubBytes();
        ShiftRows();
        AddRoundKey(key);

        printf("\nCiphertext:\n");
        printState();

        return 0;
    }
```

Output:

```
Enter 16 plaintext values:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Enter 16 key values:
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1

Ciphertext:
14 7 2 2
3 11 8 9
0 4 11 2
14 1 4 5
```